

Applications of binomial distribution



- 1
 - a $\frac{125}{3888}$
 - b $\frac{625}{648}$
 - c $\frac{3875}{3888}$
- 2 0.02
- 3 0.0686
- 4 0.0839
- 5 0.2446
- 6 0.0002
- 7
 - a 0.003318
 - b 0.0333
 - c 0.1996
- 8 0.0705
- 9 0.8750
- 10 0.1356
- 11 5.7×10^{-11}
- 12
 - a Binomial, 0.4 is the probability of "success" - in this case, selecting a dinosaur to play with.
 - b
 - i 0.161
 - ii 0.071
 - iii 0.930
- 13 70
- 14
 - a 7.5 days
 - b The chance of rain on any given day is not independent of the weather on previous days.
- 15 $n = 8$

16

a 12

b 9



c 3

17

a 15

b 12

18

a

i 8

ii 8

iii 2.53

iv 2.65

b X

c 0.68

d 0.66

19

a 13.72

b 9.9

20

a 720

b 25.17

21

a $n = 25, p = 0.2$

b 0.36

22

a 0.2363

b 0.9102

c 0.7338

d 0.9630

e 0.0001

23

a 0.00047

b 0.11806

c 0.02449

d 0.21321

24

a 8

b $\frac{4\sqrt{10}}{5}$

c $P(3 \leq X \leq 13) = 0.97$

25

a 0.85

b 0.15

c $1 - (0.98)^n$

d 149

e $Y \sim B(12, 0.15)$, hence $P(Y = y) = \binom{12}{y} \times 0.15^y \times 0.85^{12-y}$

f 0.91

26

a 0.0256



b 1

27

a $p = \frac{3}{8}$

b 0.86

c $P(X = x) = \binom{12}{x} \times 0.86^x \times 0.14^{12-x}$

d 0.29

28

a

i 0.0235

ii 0.0775

iii 0.0040

b Yes, the event is unlikely to happen by chance. It is more likely there is a problem in the kitchen.

29

a 0.0197

b 0.0207

c 0.7977

d 0.9870

30

a 0.0001

b 0.0198

c 1.3×10^{-22}

d 0.9967

e 0.9997

31

a 0.0006

b 0.0250

c $n = 6$

32

a 0.2

b 0.9895

c 0.2208

d $n = 21$

33

a 0.00047

b 0.0117

c 1.1×10^{-25}

34

a 0.0115

b 0.0069

c 0.0188

35

a $\frac{25}{1296}$

b $\frac{25}{216}$

c $X \sim B\left(10, \frac{25}{216}\right)$, hence $P(X = x) = \binom{10}{x} \times \left(\frac{25}{216}\right)^x \times \left(\frac{191}{216}\right)^{10-x}$, for $x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ and 10.

d 0.90



36

a ${}_nC_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{n-3}$ for $n = 3, 4, 5, \dots$

b 10

37

a 0.1664

b 0.2717

c 8.790×10^{-7}

d 12

e 11.7

f 0.01294

g 0.1006